Q-TRACE

Quantum-Classical Temporal Risk Attribution for Trustworthy Agentic Cloud Security

PI	Professor Dr. Hafiz Md. Hasan Babu, Department of Computer Science and Engineering, University of Dhaka, Bangladesh (hafizbabu@du.ac.bd)
Co-PI	Dr. Moinul Islam Zaber, Research Fellow, Institute of Development Studies, University of Sussex; Visiting Academic, Department of Computer Science and Technology, University of Cambridge, UK (miz21@cam.ac.uk)
Co-PI	Sajib Kumar Mitra, Joint Director (ICT), Information Systems Audit Division, Internal Audit Department, Bangladesh Bank, Bangladesh (sajib.mitra@bb.org.bd)
Cash funding needed	USD 64,000
AWS Promotional Credits needed	USD 28,000
Amazon contact (optional)	Not identified

Abstract

Tool-using AI agents combine language-model reasoning with cloud identities, APIs, and sensitive data. Indirect prompt injection or a malicious tool response can therefore redirect a benign task into a multi-step chain of privilege misuse, unauthorized access, or disclosure—a pattern that flat event classifiers routinely miss. Q-TRACE will create an auditable detection framework built around temporal authorization-and-provenance graphs. A deployable classical graph/sequence model will be the primary baseline; compact quantum-kernel and variational models will test the narrower hypothesis that quantum feature maps improve rare-threat ranking under limited and noisy labels. Equal-budget classical, random-feature, and ablation controls will isolate any quantum-specific effect. Each alert will carry a policy-linked evidence trail and non-destructive counterfactual, and a distilled classical monitor will support low-latency replay on AWS. Outputs will include a safe benchmark of 8,000–12,000 traces, open-source models and evaluation tooling, an operational cost report, and a peer-reviewed manuscript.

Keywords: agentic AI security; cloud threat detection; temporal heterogeneous graphs; prompt injection; access governance; quantum machine learning; explainable AI

Motivation, significance, and novelty

The Fall 2026 call prioritizes trustworthy security agents, access governance, cloud threat detection, scalable graph modeling, and learning from limited or noisy labels [1]. In tool-using agents, untrusted content can steer a sequence of actions, and delegated permissions turn a reasoning failure into a confused-deputy event. OWASP and AWS guidance therefore emphasize prompt injection, disclosure, excessive agency, scoped authorization, observability, and layered controls [2,3]. AgentDojo exposes important attacks [4] but does not provide a cloud-oriented trace of actors, authorization, resources, and provenance. Q-TRACE contributes (i) a versioned temporal event graph and leakage-resistant benchmark; (ii) calibrated detection that joins learned signals with deterministic policy checks; and (iii) a preregistered attribution audit that prevents classical preprocessing or capacity from being mislabeled as a quantum advantage [5].

Research questions

RQ1. Do temporal authorization-and-provenance graphs outperform flat events on unseen attack families, tools, and workload conditions?

RQ2. Under identical features and search budgets, do compact quantum models improve rare-

threat ranking, recall at low false-positive rates, or calibration beyond matched classical and random-feature controls?

RQ3. Can policy-linked evidence and conservative counterfactuals make alerts more actionable without weakening detection?

Methods and technical approach

Q-TRACE treats each run as a typed, time-ordered security object. Figure 1 shows the shared representation, controlled comparison, and evidence-linked output.

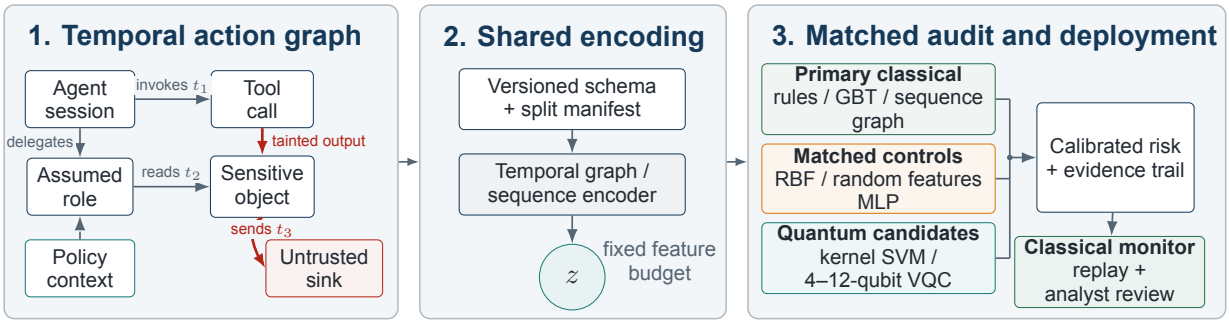


Figure 1: Q-TRACE architecture. One embedded path is scored by classical baselines, matched controls, and compact quantum candidates; deployment remains classical.

WP1—Trace benchmark and safe testbed (Months 1–4). A versioned schema will record time, task, action, principal/role, permission, tool, resource, provenance, outcome, and policy. Licensed public tasks and isolated sandbox workflows will cover prompt injection, malicious tool output, excessive agency, scope escalation, suspicious cross-resource access, and attempted exfiltration, with benign exceptions, drift, and label noise. The target is 8,000–12,000 synthetic or legally reusable traces with provenance, label confidence, and immutable split manifests. No production credentials, personal data, live targets, or exploit payloads will be collected.

WP2—Temporal graph detector and classical baselines (Months 2–6). Nodes are sessions, identities, tools, resources, data objects, and policies; timestamped edges encode invocation, delegation, access, and derivation. Features capture rarity, timing, permission transitions, trust-boundary crossings, provenance, and task deviation. We will compare calibrated policy rules, gradient-boosted trees, a sequence encoder, and a temporal graph model. Grouped partitions will keep near-duplicates and attack templates out of training.

WP3—Quantum hypotheses and attribution audit (Months 4–8). A graph encoder trained without test access will emit a fixed-budget vector z. The same z, split, seeds, and budget will train a quantum-kernel SVM, a 4–12-qubit variational classifier, matched classical controls (RBF SVM, random features, MLP), and quantum-layer ablations. Noise-swept simulations, and hardware runs if accessible, will be reported separately; the audit will isolate the quantum map from preprocessing and unequal optimization.

Evaluation, explanation, and deployment

WP4—Evidence-linked alerts and efficient deployment (Months 7–11). Each alert will name influential actions, role/tool transitions, policy mismatches, and provenance edges. Counterfactuals will note minimal observed changes linked to lower risk (for example, a narrower role or dropping an untrusted data edge) and will not claim intent. Policy gates may request analyst review or sandbox containment; the prototype will not disable production accounts or perform destructive remediation. If a research model is too slow, calibrated scores and selected features will be distilled into a compact classical monitor.

Leakage-resistant evaluation

Splits will be grouped by template and near-duplicate cluster, with held-out tests by attack family, tool, and workload. Model selection will use only training and validation data. The primary metric is AUPRC; secondary metrics are recall at a 1% false-positive rate, false alerts per 1,000 benign traces, calibration (Brier score, expected calibration error), 95th-percentile latency, throughput, memory, and cost per million events. We will use at least five seeds, paired 95% grouped-bootstrap intervals, Benjamini–Hochberg correction on preregistered contrasts, and sensitivity analyses over label noise, prevalence, graph size, and simulated device noise.

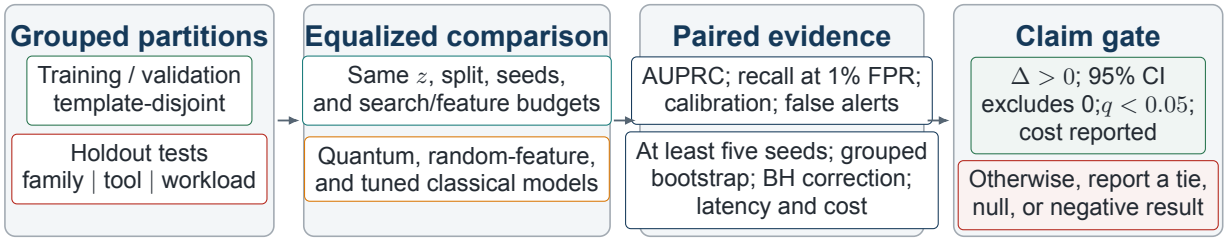


Figure 2: Falsifiable evaluation and claim gate. Shared representations and budgets; a quantum claim needs statistical, operational, and reproducibility evidence, else the result is a tie or null.

Preregistered success criteria

- **Benchmark quality:** 8,000–12,000 documented traces; zero template or near-duplicate leakage across released partitions; reproducible generation and checks.
- **Security utility:** an absolute AUPRC gain of at least five percentage points for the temporal representation over the strongest flat baseline on at least two held-out shifts, with performance also reported at a 1% false-positive operating point.
- **Quantum claim gate:** a benefit will be claimed only when the matched quantum model has a positive effect on the preregistered primary contrast, its 95% interval excludes zero, the result survives multiplicity correction ($q < 0.05$), and compute cost is fully reported. No advantage claim is required for project success.
- **Operational artifact:** replayable AWS pipeline, evidence retrieval, model card, threat-model card, and a classical monitor with measured latency and cost.

Risk management and responsible release

If QML adds no value, the benchmark, classical detector, and attribution audit remain complete contributions. Non-redistributable data will be replaced by manifests and generators; unavailable hardware will use the preregistered simulator-only path. Distribution shift will be measured rather than hidden. Releases will stay on synthetic resources, omit secrets and harmful operational detail, distinguish anomalies from incidents, and document limitations and intended use.

Months	Milestone	Measurable output
1–2	Protocol and schema	Threat model, schema, sandbox, preregistration, leakage tests
3–4	Benchmark v0.1	Safe scenarios, target of 8,000–12,000 traces, label report, split manifests
5–7	Models and audit	Graph/sequence baselines; kernel/VQC; matched controls and ablations
8–9	Robustness and evidence	Held-out shifts; calibrated evidence and counterfactuals
10–11	Deployment and release candidate	Distilled monitor; AWS latency/cost study; release candidate; draft paper
12	Public release	Version 1.0 code and benchmark, final report, cards, manuscript

Expected results, milestones, and impact

Deliverables are (D1) a versioned trace benchmark and leakage audit; (D2) graph, sequence, policy, quantum, and matched-control implementations; (D3) evidence-linked alerts and a distilled monitor; and (D4) an AWS cost report, reproducibility package, model and threat cards, technical report, and manuscript submitted to a peer-reviewed venue.

Funds needed and justification

Requested: USD 64,000 cash and USD 28,000 AWS Promotional Credits; the awards panel will determine the final amounts. No administrative or indirect overhead is requested. Cash will support research labor, benchmark engineering, secure testbed needs, independent evaluation, open-source release, and dissemination.

Direct cash costs	USD
Graduate research assistant (12 months, part-time)	24,000
Two part-time student research assistants	12,000
Benchmark engineering, data validation and reproducibility support	8,000
Local secure testbed, storage and essential non-AWS equipment	7,000
Independent security evaluation and review support	4,000
Open-source packaging, documentation, and dissemination	4,000
Publication and dissemination travel	5,000
TOTAL direct cash requested	**64,000**

Estimated AWS credits: EC2 CPU/GPU, $11,000; S3, $2,500; SageMaker, $5,000; CloudWatch and OpenSearch, $4,500; Bedrock, $3,000; and Braket, $2,000. Total: USD 28,000. If quantum hardware is unavailable or ineligible, the Braket allocation will be redirected to approved simulation and evaluation workloads.

Additional information and call alignment

Q-TRACE addresses trustworthy agentic security operations, access governance, cloud anomaly detection, toolchain security, graph modeling, and weak supervision. Planned outputs are Apache-2.0 code and, where permitted, a CC BY 4.0 benchmark with tutorials, fixed seeds, tests, and limitations. No Amazon data, endorsement, or proprietary access is assumed; quantum null findings remain explicit deliverables.

References

[1] Amazon Science. “AI for Information Security call for proposals—Fall 2026.” *Amazon Research Awards*, 2026. https://www.amazon.science/research-awards/call-for-proposals/ai-for-information-security-call-for-proposals-fall-2026

[2] OWASP GenAI Security Project. “2025 Top 10 Risk & Mitigations for LLMs and Gen AI Apps.” 2025. https://genai.owasp.org/llm-top-10/

[3] Amazon Web Services. “Mapping to OWASP Top 10 for LLM applications.” *AWS Prescriptive Guidance*. https://docs.aws.amazon.com/prescriptive-guidance/latest/agentic-ai-security/owasp-top-ten.html

[4] E. Debenedetti, J. Zhang, M. Balunović, L. Beurer-Kellner, M. Fischer, and F. Tramèr. “AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents.” *Advances in Neural Information Processing Systems 37, Datasets and Benchmarks Track*, 2024. https://proceedings.neurips.cc/paper_files/paper/2024/hash/97091a5177d8dc64b1da8bf3e1f6fb54-Abstract-Datasets_and_Benchmarks_Track.html

[5] S. A. Gillani, M. S. A. Baig, S. Munir Shah, A. Ali, and H. Siddiqui. “How Quantum Is the Advantage? A Fair, Calibration- and Noise-Aware Benchmark and Attribution Audit of Quantum Machine Learning for Network Intrusion Detection.” *arXiv:2608.18155*, 2026. https://arxiv.org/abs/2608.18155